Sub-exponential ECDLP over prime fields: a desk evaluation against the lead atlas

Date: 2026-11-01. **Study:** `research/ecdlp_subexp_desk_20261101/` (README with the full tables, data, figure source). **Atlas:** `aburan28/crypto-autoresearcher main` cfc1dae2, `knowledge/leads/ecdlp/` (154 leads, reviewed 2026-10-30). **Classification:** desk evaluation. No improvement is claimed, nothing was measured, no disposition changes.

1. Why every sub-exponential route must read the encoding, and where each one stops

Shoup's bound (`LEAD-RHO-generic-sqrt-lower-bound`) puts every algorithm that uses only the group law and equality at $\Omega\left(\sqrt{N}\right)$ on a prime-order subgroup, and at $S \cdot T^2 \gtrsim N$ with advice S. A sub-exponential algorithm therefore reads coordinates, the curve equation or the field. The atlas sorts the program's work by the routes through which an encoding can be read; for prime fields each route ends at a named obstruction (lead ids in the README, section 1):

- **algebraic factor bases** (Semaev systems): a size-B algebraic set over $\mathbb{F}_p$ has description degree $\geq B/3$, and the mean decomposition yield is $\binom{B+m-1}{m}/N$ for every base of size B (theorem; measured ratio 1.000000 to a random base);
- **small-root decomposition** (Coppersmith): every lattice on a convex monomial set reaches at most 0.3293 per S_3 block against a demand of 0.8889 (proof);
- **representations and subset sums**: generic, so $T \gtrsim \sqrt{N}$ (theorem); **enumerative, pair-table and MITM oracles**: no sub-rho cell at any arity (cost model);
- **p-adic lifts**: prime-to-p torsion lifts uniquely, so every additive order-ℓ image is zero; jet oracles are GGM-simulable (proof); the anomalous channel needs $r = p$;
- **multiplicative transports**: a homomorphic image factors through the embedding degree; no self-pairing on a prime-order subgroup;
- **covers and Jacobians over the same field**: index calculus costs $p^{2-2/g} \geq p$ against rho's $p^{1/2}$;
- **the isogeny class and endomorphisms**: generic cost is an isogeny-class invariant; $\sqrt{|\mathrm{Aut}|}$ at most;
- **global lifts**: the $\mathbb{Q}$-lift (xedni) is closed by the literature; the function-field section timing is open.

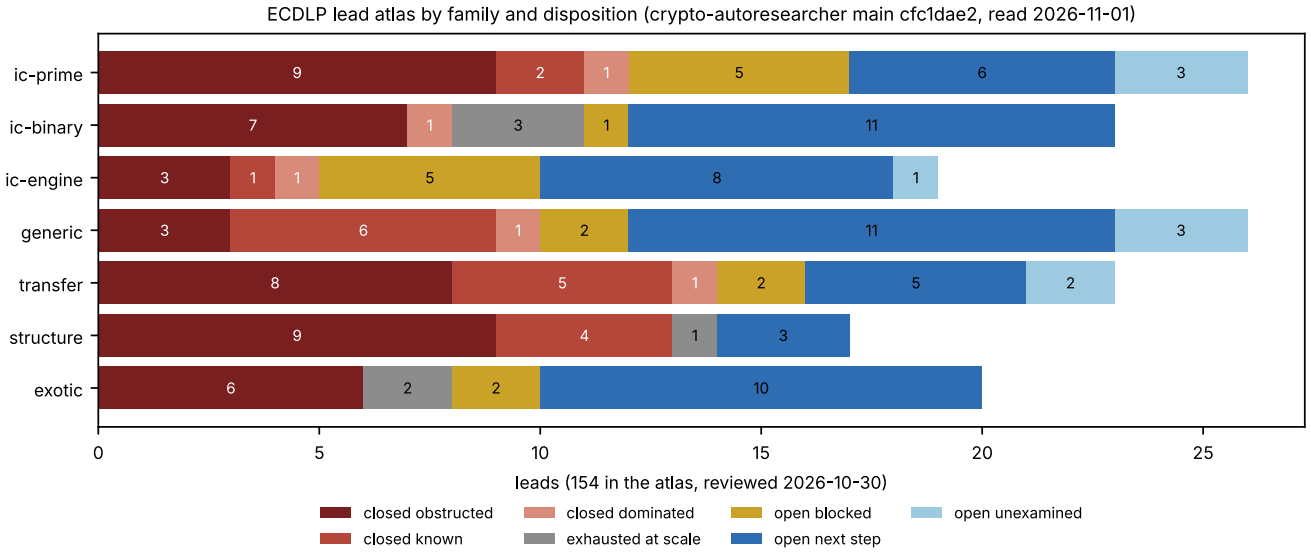


Figure 1: The atlas by family and disposition, as read from the 154 lead files (`data/atlas_tally.json`). 74 leads are closed with a named obstruction, 54 have a decided next step, 17 are blocked, 9 are unexamined.

2. Sixteen candidate mechanisms, each against the atlas

The sixteen candidates were drawn from the standard cryptanalytic toolbox and from the routes above, and each was put through `python3 tools/lead_atlas.py --match`. Every one lands on an existing lead (figure 2; raw hits in `data/candidate_matches.json`; the reading of each hit is in the README, section 2). Ten land on closed or exhausted leads, two on blocked ones, four on open ones (`LEAD-RHO-cheaper-step-primitives`, `LEAD-EXO-invertible-hint-subset`, `LEAD-EXO-four-list-ktree-join`, `LEAD-STR-class-group-ecdlp-bridge`), and none is new. The atlas's rule for this case is to advance the matched lead's next step, not to file a duplicate.

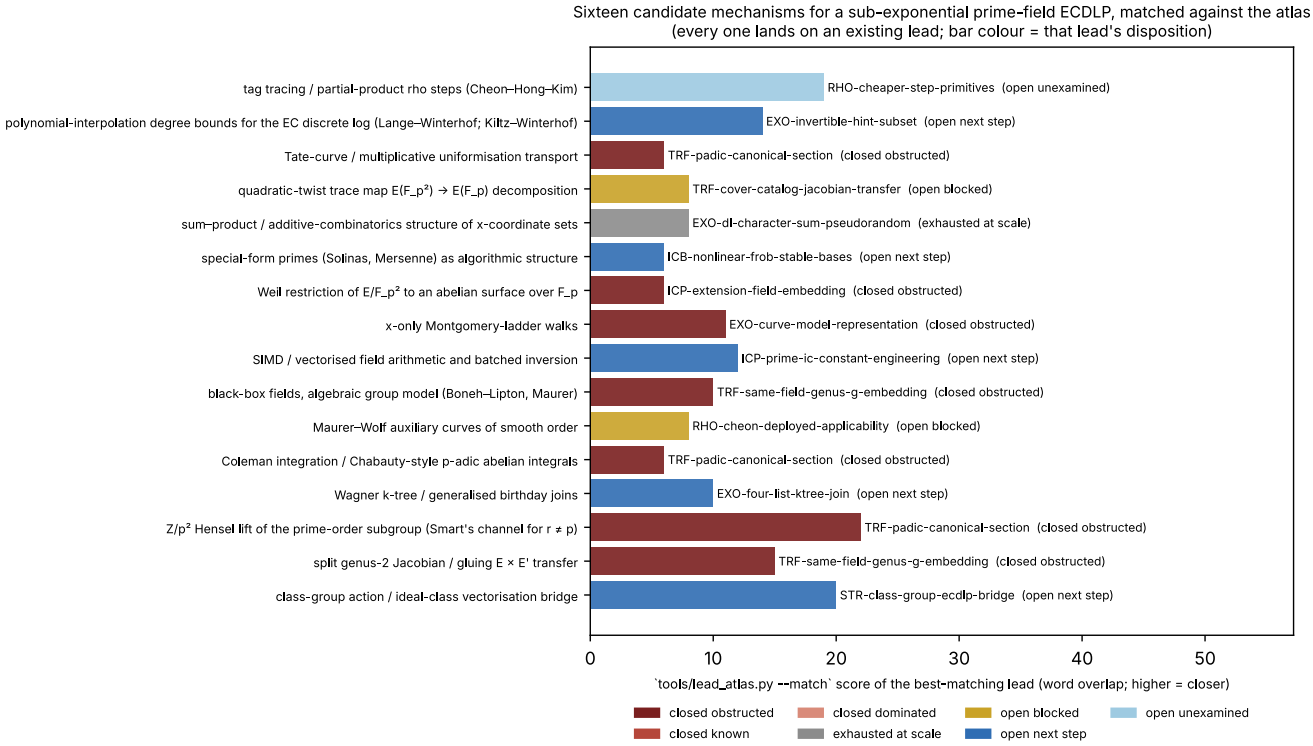


Figure 2: Each candidate's best `--match` hit (word-overlap score) and that lead's disposition. The "best lead" is the author's reading of the hits; the score is not a verdict.

One pointer the atlas lacks: the polynomial-interpolation degree lower bounds for the elliptic-curve discrete logarithm (Lange and Winterhof, "Interpolation of the elliptic curve Diffie-Hellman mapping", 2003; "On degrees of polynomial interpolations related to elliptic curve cryptography", WCC 2005) bear on the open next step of `LEAD-EXO-invertible-hint-subset`, which asks for the maximal agreement of degree-d polynomials with the log graph. The theorem statements were not retrieved in this session; the pointer is recorded as unverified.

3. What a "drastic change" can honestly mean

1. On prime fields the exponent is $1/2$ on every route the program has measured; the routes that could lower it are the open algebraic ones, and each has a cheap, decisive next step (section 4). None has a measured excess over its matched null.
2. Below-$\sqrt{N}$ exponents are known on extension fields $E\left(\mathbb{F}_{p^n}\right)$, $n = 3, 5$ (Gaudry, Diem, Joux–Vitse); the program's open leads there are the only places where such an exponent is a measurement away. Moving effort there changes the target, and is not a prime-field result.
3. The owner's criterion (AGENTS.md, 2026-10-15) counts every measured improvement over the current index-calculus baseline. The prime-field baseline last moved on 2026-10-29 ($N = 45$, $0.818[0.773, 0.866]$ at $w = 0$, 20 bits); the relation phase is 68–93% of S on Table B. The distance to rho on that table is 5.5–13.5× at 10–24 bits.

4. Open next steps, ranked by cost and by what each outcome decides

lead	next step (atlas)	cost	decides
`ICP-digit-presentation-solving-degree`	certified degree-growth ladder, $m = 2, 3$, $s = 2..8$, independent primes, matched nulls	toy, CPU-hours	semi-regular growth closes the lane at scale; a bounded d_{lf} below the null is the first elliptic early fall over $\mathbb{F}_p$
`STR-class-group-ecdlp-bridge`	the simulation lemma: curve-aware GGM with free class-group action still needs $\Omega\left(\sqrt{N}\right)$	desk / Lean	a checked simulator closes the lane with a theorem; a failure names the one bridge a reduction must use
`EXO-invertible-hint-subset`	the interpolation-degree bounds first, then $A(d)$ on the toy ladder	desk, then toy	bounds forcing degree $\geq c\lvert S\rvert$ close the low-degree variant as known
`TRF-miller-cocycle-nonhom-transfer` / `EXO-miller-pairing-cocycle`	verify K1/K2 on toy curves; equidistribution of $\ell(k)$ with a planted control	toy, hours	equidistribution at the $1/p$ rate closes both

`EXO-four-list-ktree-join`	the bucket-variance class W of the O2 kernel against $1/\sqrt{N}$	toy, CPU-hours	slower decay reopens a bucket-concentrated filter
`EXO-global-lift-height`	$T_{\mathrm{section}(d,\chi,p)}$ on E_1 surfaces, ≥ 3 primes, Gröbner control	CPU-days	$T > p^{1/2}\,\mathrm{poly}(\log p)$ closes the function-field variant
`RHO-cheaper-step-primitives`	tag-tracing walk on a 64-bit toy curve, ns per useful step	engineering, hours	< 0.95 opens an engineering lead; ≥ 1.0 closes it

5. Scope

This note proposes no new lead, measures nothing and changes no disposition. The candidates are one author's choice; `--match` is a word-overlap score and the "best lead" column is a reading of its hits. Python here only lays out recorded values (`plot.py`); the data files are the atlas's own YAML, re-serialised.